

Database System Proposal

Fleet Management System Database

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Organization Overview

The purpose of this proposal is to develop a relational database model for the Fleet Management System. A relational database model is a simple but highly effective way to organize data into two-dimensional tables (Stair & Reynolds, 2018). Prior to the development of the relational database model, organizations used navigational or hierarchical models, which made the retrieval of data complex and dependent on the underlying structure (IBM, 2021). Without a clear relational structure, data often had to be stored in multiple places, leading to unnecessary duplication and inconsistencies across databases.

In 1970, Edgar Codd created a solution to address some of these challenges when he published *A Relational Model of Data for Large Shared Data Banks* (Codd, 1970). This method makes for simpler data retrievals and minimizes unnecessary redundancies of data. It also improves adaptability and data independence allowing businesses to change how data is stored without affecting the entire system. Additionally, relational database models help enforce consistency and accuracy automatically, protecting the business from data entry errors and conflicting records (Codd, 1970).

By implementing a relational database model, the Fleet Management System will be able to produce an asset listing of all vehicles in the company's fleet including information on the trucks, acquisition costs, depreciation, operational costs, and maintenance costs. This system will help streamline fleet tracking, ensure accurate financial reporting, and enhance operational efficiencies by identifying cost trends.

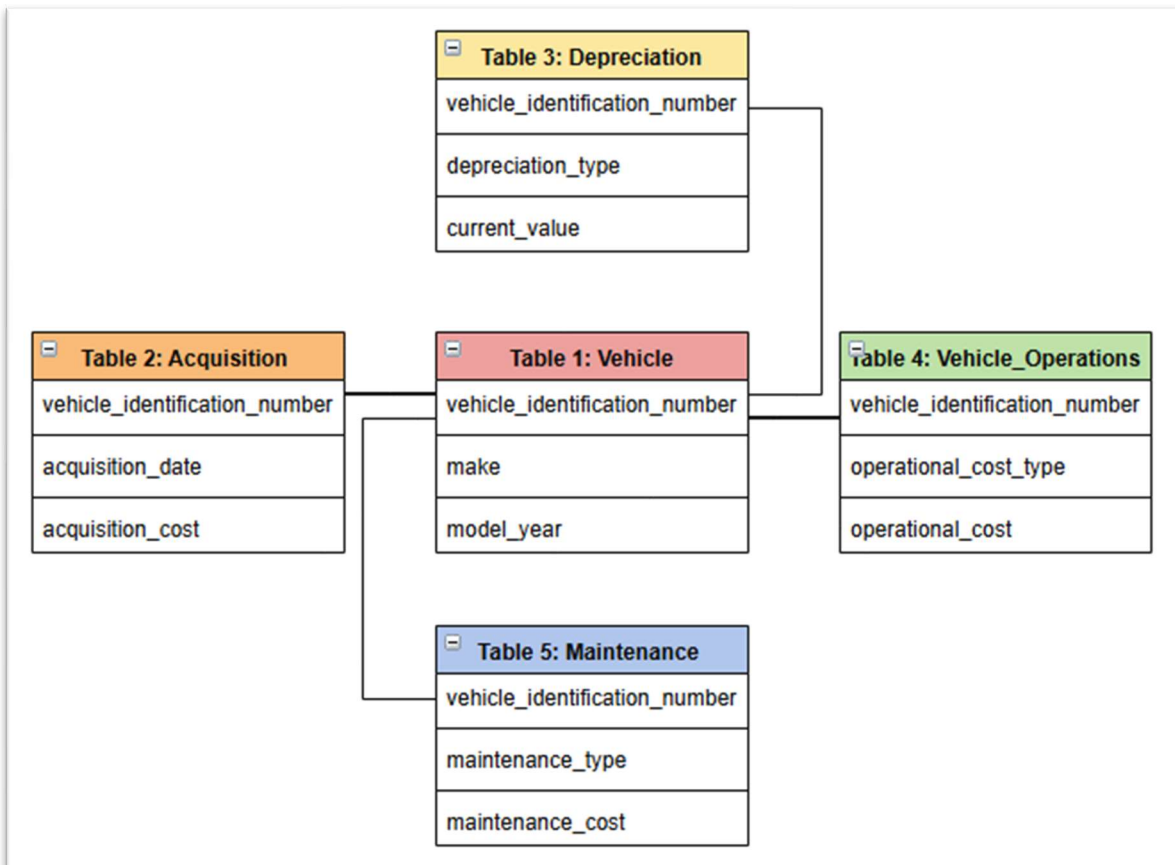
Relational Database Model Schema

Figure 1 is a list of tables in this model for use in the Fleet Management Systems. The schema contains the top three attributes related to the firm's vehicle assets for each table. The tables can be linked together by the Vehicle Identification Number (VIN) as the primary and foreign keys. This is a unique identifier that will only correlate with a specific vehicle. It also cannot be changed, unlike the vehicles' license plate numbers.

In Figure 1, Table 1: Vehicle forms the core of the model. It provides unique identifiers and basic information on the vehicles including VIN, make, model, license plate number, and year. Table 2: Acquisition ensures proper cost tracking for the acquisition of the vehicles. It includes the VIN as its foreign key linking it to other tables in the schema. The acquisition data and cost are important to include for financial reporting. Table 3: Depreciation is crucial for accounting and resale purposes. The depreciation type includes mileage, age, accidents, and other ways a vehicle can depreciate. These affect the worth of the vehicles, so it is important to include the current value. Table 4: Vehicle_Operations and Table 5: Maintenance helps track ongoing costs, enabling better decision-making regarding cost-efficiencies. Operating Expenses (OPEX) refers to the regular costs a business incurs to maintain its day-to-day operations (Forage, 2023). For the Fleet Management systems, the types of costs will include fuel, fluids, insurance, and registration. Maintenance costs are expenses organizations have to keep the vehicles in working condition (Garibian, n.d.). For vehicles, this includes repairs, oil changes, brake replacement, and other tune-up maintenance. In summary, the Fleet Management Systems database is structured around core tables that capture essential vehicle information, cost, and type of cost providing a framework for accurate tracking, reporting, and cost management.

Figure 1

Relational Database Model for the Fleet Management System



Note. To support the Fleet Management System, this relational database schema will be implemented. This figure provides a list of tables with their top three attributes.

Application Areas

Application areas play a vital role in utilizing the Fleet Management System and keeping it updated. Relational database models require application areas for transactional data where ACID (Atomicity, Consistency, Isolation, Durability) properties are crucial. ACID-compliant databases offer zero loss of data during failures, no accidental creation of inaccurate or duplicate data, and consistency, predictability, and reliability in operations. It allows businesses to make better decisions, minimize issues caused by error, and operate smoother (YugabyteDB, n.d.). ACID properties ensure database transitions are executed reliably and make certain of the integrity of data in the database (Stair & Reynolds, 2018). Figure 2 outlines some of the departments who will contribute to the Fleet Management System and how the tables apply to their functions. It also provides guidance on what transactional data the departments are responsible for updating.

Figure 2*Applicational Areas for Transactional Data in the Fleet Management System*

Purchasing	• Tracks vehicle acquisitions, supplier information, purchase orders, and acquisition costs • Updates vehicle inventory with new vehicle transitions
Finance & Accounting	• Manages depreciation for vehicles • Tracks financial impacts of expenses, including repairs, fuel, and insurance • Generates financial reports related to cost and budget compliance
Operations	• Logs daily operational expenses, including fuel consumption and mile usage
Maintenance & Repairs	• Tracks vehicle maintenance and repairs for schedule and unscheduled services • Updates maintenance records, service history, and schedules future vehicle servings
Accounts Payable	• Manages vendor payments for fuel, maintenance, repairs, and other services • Updates financial records for operational and maintenance costs
Scheduling	• Utilizes reports to allocate vehicles to assignments • Updates the vehicles' status (available, in-use, in-maintenance), vehicle location, and optimizes fleet usage

Note. This figure illustrates the key application areas within the Fleet Management system that generate transactional data.

Each department contributes unique data that is essential for maintaining accurate and up-to-date records. Purchasing ensures accurate records of newly acquired vehicles, while Finance & Accounting integrates depreciation for financial tracking. Operations submits monitoring of fuel and operational expenses. The Maintenance and Repairs Department is essential for tracking repair history and scheduling maintenance. Accounts Payable handles the company's short-term financial obligations and updates the records once outstanding costs have been paid. Scheduling is essential for vehicle allocation. Together, these departments ensure that the Fleet Management System operates efficiently and adheres to ACID properties.

System Diagram

A systems diagram displays the interactions, effects, and processes of a specific system. The goal is to be easily understandable and contain accurate and relevant information (Indeed Editorial Team, 2021). They are more than just flow charts. It provides teams with a clear, complete, and common understanding of a system. “The main enemy of system architects and software engineers is complexity. Complexity leads to a poor understanding of a system, which in turn can make the task of modifying or even using the system very challenging” (Scala, 2019). The system diagram is designed to provide a simple method to solve this complex scenario. Figure 3 is a system diagram for the Fleet Management Systems that includes databases, source data applications, and the application area. These applications act as data sources that supply important datasets into the Fleet Management System. The system

databases, implement changes to improve performance, and guarantee that the data is secure from unauthorized access (Stair & Reynolds, 2018).

The implementation of a relational database model for the Fleet Management System provides an opportunity to streamline fleet tracking, ensure accurate financial reporting, and enhance operational efficiencies by identifying cost trends. Using this model, the organization will be able to produce an asset listing of all vehicles in the company's fleet including information on the trucks, acquisition costs, depreciation, operational costs, and maintenance costs. They will be able to address the problem of unnecessary duplication and inconsistencies across databases. By implementing this relational database model, the company will be able to make more informed decisions, reduce costs, and enhance the efficiency of its fleet management operations.

References

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